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Impact of a school-based nutrition educational intervention on knowledge related to iron deficiency anaemia in rural Karnataka, India: A mixed methods pre-post interventional study

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Abstract

Objective: To understand the extent to which adolescent awareness of and attitudes about anaemia and anaemia prevention can be changed by nutrition messages received at school. **Design:** Mixed-methods pre–post intervention study Setting: Three Government schools in Bagalkot, Belagavi and Raichur districts of Karnataka, India **Population or Sample:** Students of grade six and seven and teachers involved in implementing the intervention. **Methods:** An education intervention was co-developed by school teachers and nutrition experts using locally adapted resource materials that consisted of lectures, role play and practical demonstrations. Seven half-hour educational sessions were delivered by school teachers over seven weeks to 455 students. Pre- and post-intervention tests measured changes in adolescents' knowledge about anaemia. In-depth interviews with teachers and focus groups with students explored reactions to the intervention. **Main outcome measures:** Knowledge score related to anaemia Results: The percentage of children with correct scores increased by 7.3 - 49.0 percentage points for the tested questions after implementation of the intervention. The mean knowledge score increased by 3.67 ± 0.17 ($p<0.01$). During interviews, teachers and students highlighted high acceptance of the intervention and materials, an increase in awareness, a positive attitude towards changing behaviour around diet, an increase in the demand for iron folic acid supplements and improved sharing of messages learned with peers and families. Challenges expressed included need for further training, time limitations and hesitancy in teaching about menstruation and pregnancy. **Conclusions:** Educational interventions carried out for adolescents by teachers in schools are effective in improving awareness and attitude related to anaemia.

Introduction

Despite several decades of programming, anaemia continues to be a critical public health problem in India, affecting 59% adolescent girls and 31% boys aged 15-19 years.[1] This is an increase from the 2015-16 estimates and has raised concerns regarding the effectiveness of anaemia prevention programs.[2, 3]In India, the prevalence of anaemia is highest in the early years, decreases until about 11 years of age and then increases

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again during adolescence, particularly in girls due to the start of menstruation and as a consequence of early marriage and teenage pregnancy.[4-8] Nutritional anaemia due to lack of iron, folate, or vitamin B12 is the most prevalent form in this age group[6] and has long term negative implications on growth and development, school performance and work productivity.[8-12] Anaemia during adolescent pregnancy is associated with poor pregnancy outcomes such premature births, low birth weight, and perinatal and maternal mortality.[8, 13, 14]

The Government of India has highlighted the importance of improving the nutrition of adolescents by modifying schemes originally aimed at the welfare of pregnant women and children. The Scheme for Adolescent Girls (SAG), the *Kishori Shakti Yojana* , RMNCH+A initiative, Sneha Clinics, the National Iron Plus Initiative, the Anaemia *Mukt Bharat* (AMB, Anaemia free India) initiative and *POSHAN* Abhiyaan, all include special features for empowering and improving the health and nutrition of adolescents. The revamped AMB initiative, aimed to break the intergenerational cycle of anaemia, focuses on six groups likely to benefit, six institutional mechanisms and six interventions.[15, 16] Several of these interventions such as weekly iron and folic acid (IFA) supplementation, anaemia testing and treatment, deworming and education, specifically target adolescent boys and girls.[15]

Regardless of these interventions, findings from our recent study in rural Karnataka and others studies as well, revealed gaps in community awareness of adolescent anaemia and its prevention, poor dietary behaviour, low compliance with regular IFA consumption and lack of adequate nutrition education in schools.[17-21] Students and teachers described a vigilance-oriented approach i.e., supervised swallowing of IFA tablets, without any meaningful communication on what they were receiving, and why.[22] In addition to addressing supply side barriers and careful programming, studies have continually highlighted a need to implement interventions to address demand-side barriers by improving awareness and changing behaviours surrounding diet and iron consumption.[4, 16, 19, 23-27]To this end, we conducted research to understand the extent to which adolescent awareness of and attitudes about anaemia can change through short, targeted nutritional messages delivered by teachers at school.

Material and Methods

Design and setting

To investigate the effects of targeted nutritional education, a mixed-methods pre–post intervention study was conducted in three Government schools of Karnataka, one each in Bagalkot, Belagavi and Raichur districts. All students in grades six and seven in the three schools were invited to participate in the study. Written informed consent was obtained from the students' parents and assent from the students. Pre and post - intervention tests were used to measure changes in adolescents' awareness due to the intervention, and qualitative interviews were used to capture reactions and facilitators or barriers to the implementation of the intervention.

Nutrition education intervention

A multimodal learning approach was used as a basis to develop the intervention. The education intervention was delivered through teachers and divided into seven sessions containing lectures, role play and practical demonstrations. Teachers were chosen as educators to ensure sustainability of the intervention and students of grades six and seven were chosen, considering the higher attrition among older students. The content and modes of delivery were co-developed through multiple deliberations and discussions with community leaders, local school teachers and nutrition experts. The resource materials were adapted from the resources available for the Anemia Mukt Bharat programme and the National Institute of Nutrition. The final intervention materials included illustrative, pictorial, and colourful books, cards, wall charts, handbooks for teachers and food samples (legumes, grains, dry fruits, vegetables etc.). Emphasis was put on ensuring that materials developed were appealing to children and easy to understand. All materials were translated into

the local language (Kannada) with photos of local foods added, and messages modified as appropriate for the Kannada language. The sessions covered: (i) Balanced diet; (ii) Anaemia and its importance; (iii) Anaemia in pregnancy; (iv) Diet and anaemia – iron-rich foods; (v) IFA supplementation; (vi) Myths and facts about IFA, do's / don'ts of IFA consumption; (vii) Good habits to be adopted; (viii) Summary and mantras for adolescent health. The intervention was delivered in Kannada in sessions of approximately half an hour once every week for seven weeks.

A total of 11 teachers (one male and 10 female) from the three schools were initially provided a one-day in-person training on the materials and approach before the start of the intervention. Regular online weekly refresher training was conducted during the intervention to reinforce the material and resolve any challenges.

Data collection

A pre- and post-intervention test on knowledge and attitudes about anaemia was conducted with participating adolescents before and after the education sessions. A structured pretested Knowledge Assessment Questionnaire with a maximum score of 18 was administered in Kannada. Post-intervention testing was conducted approximately 1-2 weeks after the last classroom session. In addition, post-intervention in-depth interviews (IDIs) with seven teachers and six focus group discussions (FGDs) with students, two in each district (one for girls and one for boys), were conducted to elicit information and opinions about how the intervention was carried out, including the strong and weak points of the educational methods used.

Data analysis

For quantitative data, descriptive statistics including means, standard deviations, percentages, and frequencies were used to illustrate respondents' characteristics and intervention participation. Pre- and post- knowledge scores were calculated, and paired t-tests were used to determine significant changes in knowledge scores after the implementation of the intervention. The score obtained was also categorized as low (score between 0–5), medium (score between 6–11), and high (score between 12–18). Statistical analyses were done using the statistics programme STATA 17.0 (www.stata.com). Qualitative data was analysed using an inductive thematic approach. Transcripts were read by team members to develop separate coding framework for FGDs and IDIs, based on the topic guides. Coded sections were reviewed to generate themes, and team discussions were held to cross-check findings.

Results

Background characteristics of participants

A total of 455 children were enrolled in the study, about 60% were 11-12 years of age and 56% were girls. The children were studying in year six (51.9%) or seven (48.1%) during the implementation of the intervention. A total of 413 (90.8%) children participated in both the tests: 419 (92.1%) in the pre-intervention test and 447 (98.2%) in the post-intervention test. IDIs were conducted with seven participating teachers aged 30-45 years.

Intervention compliance

Table 1 shows the attendance rates (percentages) of children in the education sessions. About 71% (86.1% in Belagavi, 64.4% in Raichur and 63.1% in Bagalkot) of the children attended a majority (6-7) of the sessions. Average attendance ranged from 5.7 to 6.3 sessions across the districts.

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[**Insert Table 1**]

Impact of the intervention on knowledge

Baseline or pre-intervention knowledge of several aspects related to anaemia, iron and iron-rich food, IFA supplements and deworming were low, with 13.1% to 46.0% of the respondents providing correct answers to the test questions. A significant increase was seen in the percentage of children who provided correct responses to the questions after the implementation of the intervention. The percentage of children with correct scores increased by 7.3-49.0 percentage points for the tested questions except for Q6_6 where most of the children did not mention "dates" as a source of iron-rich food (Table 2).

[**Insert Table 2**]

The mean score of the students in pre-intervention test was 5.64±2.66 which significantly increased to 9.32±3.52 after the intervention ($p<0.01$) (Table 3). The mean difference was 3.67±0.17.

[**Insert Table 3**]

Table 4 shows pre- and post-intervention test performance of students by district, age, gender, and attendance level. The performance of students improved in all districts, and with increasing attendance level. Although knowledge of boys was low compared to girls prior to the intervention, the gap was reduced following the intervention. No major difference was observed by age.

[**Insert Table 4**]

Perception and experience of the intervention

IDIs with teachers and FGDs with students elucidated high acceptance and positive feedback regarding the interventions, the communication materials used, the education sessions, increase in knowledge, as well as a positive attitude towards behaviour change and wider sharing of information. Teachers also highlighted some challenges in implementation and how the intervention could be improved.

Overall high acceptance and positive feedback about the intervention

Both teachers and students mentioned that the pictorial nature of the behaviour change communication (BCC) materials and the multimodal stimulation created interest in the students, including those who had difficulty in reading and writing. During the FGDs, students mentioned that they particularly liked the cards, pictorial book, and practical sessions using grains, vegetables, and other food items. These are highlighted in the excerpts below:

- *"... we have learnt a lot after reading this book. We know things in detail now; we never felt bored while reading." (Student, FGD)*
- *"... they want to see pictures and have activities; they can learn better that way." (Teacher, IDI)*
- *"Some children are not very bright, they have difficulty in reading and writing, even such children found the book very useful, they used to see the pictures and understand." (Teacher, IDI)*

Increase in knowledge

Participating teachers and students expressed that these topics were either new to them or that they did not have in-depth knowledge. Teachers perceived that their knowledge had increased after the training and after reviewing the materials, as one teacher mentioned: *"We did not even know that anaemia exists in pregnancy... Now through this syllabus, we have attained basic knowledge about why a pregnant woman has anaemia"* (Teacher, IDI). Similarly, students shared several positive influences of the education sessions on their knowledge and awareness regarding anaemia, iron consumption, anaemia in pregnancy, balanced diet etc:

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"Earlier I never used to eat well or consume tablets. After these classes, I am aware of how to eat good food." (Student, FGD)

"We learnt about parasitic infection and tablets to consume for that... they were health related. We didn't know previously, and the topics were new." (Student, FGD)

Positive attitude towards behaviour change

Although our study was not intended to measure change in behaviour, we asked teachers and students whether there were any changes after the implementation of the education sessions. Findings reflect a change in attitude and behaviour regarding vegetable consumption, exercise, handwashing, reducing junk food, tea, tobacco, alcohol consumption and an increase in the demand for IFA tablets. For example, teachers as well as students mentioned that there was more acceptance of eating vegetables after the sessions:

"... before this project began, we used to make sure that they ate all the vegetables... but now, they are doing it voluntarily." (Teacher, IDI)

"I used to not eat vegetables at school, but now I eat. After the lessons we have started eating properly. Earlier our teachers used to repeatedly tell us, but we used to ignore." (Student, FGD)

Similarly, a teacher shared an incident where a student showed her father the pictures and spoke to him about the negative effects of tobacco: *"See, if you continue to drink and chew tobacco, you will get all these diseases."* The teacher further mentioned: *"He rang me up immediately that night and thanked me. He said his daughter has "changed my mind in this manner"* (Teacher, IDI).

Increase in demand for IFA supplements

Due to Covid-19, there was a discontinuation in the supply of IFA tablets in the schools during the study period. However, several teachers and students expressed that the demand for IFA had increased after the education sessions. Teachers shared that the sessions created an understanding amongst the students that the food consumed may not be enough to meet the iron requirements and hence they were asking when the school could provide IFA tablets: *"They asked us, 'Why are we not being given the tablets?' They asked the headmistress. They have understood the concept that our food does not contain enough iron and hence we need to take the tablets. That is the reason they asked for the tablets." (Teacher, IDI)*

Sharing information with peers and relatives

Students also cited that they shared information and materials received with their family members, friends, other students who were appreciative of the topic and materials. This included informing family members about balanced meal, harmful effects of tea, tobacco and alcohol, benefits, and appropriate ways of consuming IFA tablets amongst others. One student mentioned: *"We have explained to our family members that we do not get harmed by the tablets in any way. The teachers also take them. They are supplied by the Government. Because we do not eat too many vegetables, this incorporates all that. It has lots of iron content"* (Student, FGD). Another teacher cited, "*Few girls have explained about importance of Iron and folic acid to their sisters or aunts who are pregnant... the girl students have given them correct information. We initially thought teaching people would be difficult, but it wasn't as difficult"* (Teacher, IDI).Additionally, a teacher reflected that education through children may be a better way to bring about a change in the community: *"Also people here don't cooperate much. If we force and tell them about these things, they may not accept. So, the best way to teach them is through children"* (Teacher, IDI).

Challenges in implementation

Participants, particularly teachers identified some challenges regarding training needs, time constraints and hesitancy in talking to all students together about pregnancy and menstruation.

The teachers included in the study varied by background and were not necessarily science teachers. A one-day training was arranged for all teachers, and some felt that the training duration should have been increased, whilst others expressed that, more details should have been included for teachers in the training manual.

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A few teachers (both male and female) were hesitant to discuss menstruation and pregnancy-related session to all boys and girls together and felt that the students were not interested in the topics or were more suitable for older students. As one teacher mentioned, "*... We called the 7th grade girls separately and told them about changes in body during puberty*" (Teacher, IDI). On the contrary, several teachers and students expressed that it was the topic which was most interesting.

Although all teachers expressed interest in continuing the sessions, lack of time was also consistently mentioned as a barrier. To overcome this and ensure sustainability, teachers suggested incorporating these topics as part of the primary and secondary school curriculum: "*Instead of keeping it as a separate subject, if it is incorporated in our subject textbooks, it will be better... if this happens, the basics will be covered in the primary school and as the child goes into higher classes, it will be taught more and more in depth.*" (Teacher, IDI)

Other suggestions included incorporating more activities, such as drama, plays, and videos to stimulate learning and using these methods to teach other topics.

Discussion

Main findings

The present study demonstrated that an interactive and multimodal nutrition education intervention on knowledge and attitude regarding anaemia among a sample of school-going adolescents in rural Karnataka, India is effective. We report a significant improvement in nutrition knowledge and awareness after the implementation of the intervention. IDIs with teachers and FGDs with students reveal high acceptance of the intervention and materials. Participants highlighted positive outcomes including an increase in their knowledge, a positive attitude towards behaviour change around diet and an increase in the demand for IFA supplements and dissemination of messages learnt with peers and families. Challenges expressed were related to the need for further training, time limitations, and hesitancy in talking and learning about information related to menstruation and pregnancy.

Strengths and limitations

One of the major strengths of the study was the co-development of the education materials and the implementation plan with relevant local stakeholders including teachers and health and nutrition experts to ensure acceptability and viability.[28] Based on repeated discussions with stakeholders, a few key decisions were made. These included incorporating information on balanced diet and health and hygiene in addition to topics related to anaemia, contextual adaptation of existing BCC (Behaviour Change Communication materials), implementation of the intervention among grade six and seven students, and involvement of teachers as educators.

The study was designed to measure changes in knowledge, not in behaviour. However, several studies in India have reported a change in behaviour after the implementation of school-based nutrition education interventions. Due to planned school closures because of the Covid-19 pandemic, the post-intervention test was carried out after a gap of only 1-2 weeks, and retention in knowledge was not studied. Also, to sustainably enhance knowledge, the education intervention would need to be repeatedly implemented.

Interpretation (in light of other evidence)

The findings of this study are consistent with prior studies that suggest that nutrition education in schools can improve knowledge and awareness about anaemia; increase demand and improve compliance with IFA supplements; and change adolescent dietary behaviour.[29-33] The pedagogical approach employed in the education sessions differed from conventional approaches in schools and was designed to be interactive and participatory, which we believe crucially contributed towards greater acceptance of the intervention and a positive impact on knowledge and attitude.[34] Similar studies conducted in school settings in India using

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varied approaches, such as single or multiple interactive education sessions delivered by doctors, trained experts using flipcharts, videos and other interactive methods have been found to improve knowledge and/or change behaviours related to anaemia and its prevention. [29-32]Evaluation of the Weekly Iron and Folic Acid Supplementation (WIFS) programme in India and other low-and-middle-income countries found that one of the key factors influencing increased uptake of IFA supplements was implementation of a coordinated and timely information education and communication strategy that dispelled misconceptions and rumours, and capitalised on the positive effects. [25, 35-39]Moreover, as indicated by Joe et al (2022), increased awareness and subsequent demand-generation by target groups will help address other institutional barriers related to the supply and distribution of IFA supplements.[16]

The present study also reported knowledge about anaemia to be low among boys before the start of the intervention, in concordance with several other studies.[40-42] Although recent data indicate no major gender disparities in the coverage of IFA supplementation in schools, being aware of anaemia itself acts as a motivation for consumption of IFA tablets.[17, 18] Equal emphasis must therefore be given to educating both boys and girls about anaemia and its prevention.

The inclusion of nutrition education in formal education programmes targeted to children in primary schools is one of the most commonly recommended strategies because of its potential in developing positive health, eating, and lifestyle habits that may persist into adulthood.[43, 44] Murimi et al (2018) highlighted several features of successful school-based nutrition education interventions: engagement of parents, identification of specific behaviours to be modified, ensuring fidelity by training educators, adequate dosage, and age-appropriateness of interventions.[45] Several of these features warranted more attention in the present study, including training, appropriateness of content, dosage and engagement of parents. For example, the role of teachers was essential to intervention success, but they expressed difficulties related to time constraints and training. To resolve time limitations as well to ensure continuation and viability, they suggested the content be included in the national curriculum. Gaining an understanding of anaemia motivated teachers, to help ensure fidelity and realise project objectives, but this could be further improved by providing adequate training, endorsement and implementing a teacher-friendly intervention.[25, 46-48]

Despite co-development of the content of the education sessions, some teachers were hesitant to focus on topics related to menstruation and pregnancy, particularly in the presence of boys. In patriarchal societies like India where gender norms are strict, menstruation and pregnancy are considered as 'women's' topics and exclusion of boys from related discussions can further increase gender disparities.[49, 50] Hence, research needs to be conducted to understand how best to include these topics in sessions involving both boys and girls.

One of the benefits of school-based health and nutrition education programmes is that children act to spread the information, bringing it to their homes to achieve transmission of the information to the whole family.[51, 52] Our findings reveal the same with students discussing and sharing topics learnt with friends and families. However, to achieve better impact, studies recommend involvement of families and communities in the nutritional education programmes from the beginning. [52-55]

Conclusion

Appropriate education and counselling on anaemia, iron-rich diet and IFA supplementation is crucial for adolescents and can help reduce the burden and consequences of anaemia. Our findings indicate that an education intervention employing a multimodal learning approach carried out for adolescents by teachers in schools is effective in improving awareness and attitude related to anaemia and its prevention. Adoption of such intervention in the national curriculum and scale-up may help make marked improvements in the nutritional status of adolescents. Future studies to assess impact on behaviours related to diet and consumption of IFA supplements as well anaemia status will provide insights regarding the effectiveness of the intervention.

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Disclosure of interests

None declared. Completed disclosure of interests forms are available to view online as supporting information.

Contribution to authorship

DA, GK, BG, and SS were in the conceptualisation and design of the study. DA, SG, AM, SR, BP provided overall guidance. SS and UR drafted the manuscript and incorporated all feedback. SS, UR, UC, GK, BG, JM, SP were involved in data analysis and interpretation. UR, UC, GK, JM, SP, PV, CK, AD were involved in data acquisition and overall implementation of the study. All authors read and approved the manuscript.

Details of ethics approval

The research protocol was approved by the institutional ethics committees at the three study sites and the University of Sheffield. The Health Ministry's Screening Committee at the Indian Council for Medical Research approved the project. The trial was registered with the Clinical Trial Registry of India (CTRI/2020/09/027515) on 01/09/2020.

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Table 1: Attendance in education sessions by District

Session	Bagalkot (N=149)	Bagalkot (N=149)	Belagavi (N=151)	Belagavi (N=151)	Raichur
	n	%	n	%	n
0-2 sessions	5	3.4	4	2.6	6
3-5 sessions	50	33.6	17	11.3	49
6-7 sessions	94	63.1	130	86.1	100
Mean (SD) sessions attended	149	5.67(1.24)	151	6.25(1.32)	155

Table 2: Percentage of children with correct response for each question by test type

Topic area		Questions
Anaemia	Q1	Anaemia: Low Iron (haemoglobin) in blood
	Q2	In India prevalence of anaemia is approximately 50%
	Q3_2	Anaemia can cause: Poor growth
	Q3_3	Anaemia can cause: Repeated infections
	Q3_4	Anaemia can cause: Difficulty with studies
Iron and iron rich foods	Q5	Iron consumption is mainly needed for Blood improvement
	Q6_1	The following are iron rich foods: Meat
	Q6_2	The following are iron rich foods: Sprouted vegetables
	Q6_3	The following are iron rich foods: Groundnut Laddoo
	Q6_6	The following are iron rich foods: Dates
	Q6_8	The following are iron rich foods: Green leafy Vegetables
	Q8	Guava increases the absorption of iron
	Q9	Tea inhibits the absorption of iron
IFA supplement	Q10	The Government of India recommends one IFA tablet be taken by adolescents once a wee
	Q11	The best time to consume IFA tablet is one hour after meals
	Q12	Blackish stool after IFA tablet consumption is a normal occurrence with iron consumptio
	Q13	The colour of Government supplied IFA supplement is Red/blue/pink
Deworming	Q18	Deworming is recommended once in six (6) months

Table 3: Comparison of test score before and after intervention

Score	n	Mean±2SD	Mean Difference±2SE	p-value
Pre-intervention test score	413	5.644±2.66	3.673±0.17	<0.01
Post-intervention test score	413	9.317±3.52		

Paired t-test, t, dF=-21.5, 412 SE=Standard Error, SD=Standard deviation

Table 4: Comparison of pre and post intervention performance by background characteristics

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Variables	Pre-intervention test score	Pre-intervention test score	Pre-intervention test score	Pre-intervention test score	Pre-intervention test score	Post inter-vention test score	Post inter-vention test score	Post inter-vention test score	Post inter-vention test score
	Low (0-5) (%)	**Medium (6-11) (%)**	**High (12-18) (%)**	**Mean**	**N**	**Low (0-5) (%)**	**Medium (6-11) (%)**	**High (12-18) (%)**	**Mean**
District									
Bagalkot	26.8	71.7	1.5	6.61	138	7.5	49.3	43.2	10.89
Belagavi	47.5	51.1	1.4	5.94	139	19.2	65.1	15.8	8.22
Raichur	73.2	26.8	0.0	4.35	142	13.6	66.5	20.0	8.79
Age									
11-12‘	54.7	44.9	0.4	5.25	245	13.2	62.4	24.4	9.23
13-14‘	42.0	56.3	1.7	6.14	174	13.8	57.5	28.7	9.38
Gender									
Male	53.6	44.2	2.2	5.39	181	17.8	60.4	21.8	8.73
Female	46.2	53.8	0.0	5.80	238	10.0	60.4	29.6	9.74
Intervention attendance									
Low	66.7	33.3	0.0	3.25	12	27.3	63.6	9.1	8.18
Medium	49.1	49.1	1.9	5.52	106	13.9	63.5	22.6	9.27
Regular	48.8	50.5	0.7	5.75	301	12.8	59.2	28.0	9.33
Total	49.4	49.6	1.0	5.62	419	13.4	60.4	26.2	9.29